

CPSC 4330-01/5330-01

Multimedia Processing

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Homework 1

Chapter 2 Projects and Problems

- 2.3*
- 2.7

* Note: For assignments, the questions are from the required textbook (info. given on syllabus) under the last section of each chapter, Projects and Problems. For example, 2.3 is the third questions about first-order entropy under "Chapter 2 Section 2.8 Projects and Problems" on Page 38.

2.8 Projects and Problems

1. Suppose X is a random variable that takes on values from an M -letter alphabet. Show that $0 \leq H(X) \leq \log_2 M$.
2. Show that for the case where the elements of an observed sequence are *iid*, the entropy is equal to the first-order entropy.
3. Given an alphabet $\mathcal{A} = \{a_1, a_2, a_3, a_4\}$, find the first-order entropy in the following cases:
 - (a) $P(a_1) = P(a_2) = P(a_3) = P(a_4) = \frac{1}{4}$.
 - (b) $P(a_1) = \frac{1}{2}, P(a_2) = \frac{1}{4}, P(a_3) = P(a_4) = \frac{1}{8}$.
 - (c) $P(a_1) = 0.505, P(a_2) = \frac{1}{4}, P(a_3) = \frac{1}{8},$ and $P(a_4) = 0.12$.
4. Suppose we have a source with a probability model $P = \{p_0, p_1, \dots, p_m\}$ and entropy H_P . Suppose we have another source with probability model $Q = \{q_0, q_1, \dots, q_m\}$ and entropy H_Q , where

$$q_i = p_i \quad i = 0, 1, \dots, j-2, j+1, \dots, m$$

and

7. Determine whether the following codes are uniquely decodable:

(a) $\{0, 01, 11, 111\}$

(b) $\{0, 01, 110, 111\}$

(c) $\{0, 10, 110, 111\}$

(d) $\{1, 10, 110, 111\}$

8. Using a text file compute the probabilities of each letter p_i .

(a) Assume that we need a codeword of length $\lceil \log_2 \frac{1}{p_i} \rceil$ to encode the letter i . Determine the number of bits needed to encode the file.

(b) Compute the conditional probabilities $P(i/j)$ of a letter i given that the letter is j . Assume that we need $\lceil \log_2 \frac{1}{P(i/j)} \rceil$ to represent a letter i given that the letter is j . Determine the number of bits needed to encode the file.

Requirement:

- All assignments should be typed.
- Handwritten assignments will not be accepted.
- You should have a cover page with course code, student name, date, and instructor name.